

CS486 - Design Brief I

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1 Introduction

1.1 Chosen Design Challenge

We chose the **Travel Planning** design challenge.

1.2 Project Scope

This project focuses on travel planning for short leisure trips in unfamiliar cities. More specifically, it examines how young independent, budget-conscious travelers make everyday planning decisions such as where to go, what to do, what to prioritize, and how to adapt plans based on time, budget, mood, and energy. Rather than focusing on long-term itinerary building alone, the project looks at the uncertainty and stress involved in making concrete travel decisions before and during a short trip.

1.3 Why This Problem Matters

Short leisure trips amplify the cost of poor decisions. When travelers have limited time, limited budgets, and incomplete knowledge of an unfamiliar city, the challenge is not simply finding options, but identifying which option is most suitable in the moment. Existing digital tools provide abundant information, yet research suggests that travelers still face fragmented search processes, contextual uncertainty, and information overload. This makes short-trip planning a meaningful interaction design problem rather than a purely logistical one.

2 Domain Research

2.1 Context Overview

Travel planning for short leisure trips has become highly digital, fast-moving, and information-rich. Young travelers often rely on a combination of maps, review platforms, social media, blogs, booking websites, and peer recommendations when deciding what to do in unfamiliar cities. On the one hand, this gives them access to a large amount of information and a wide variety of options. On the other hand, it creates a fragmented planning process in which relevant details are distributed across many sources rather than integrated into one coherent decision space.^{6,9}

This issue becomes especially important in the context of short urban trips. Compared with longer holidays, short trips leave less room for trial and error. A poor choice has a higher opportunity cost because travelers have fewer hours, fewer activity slots, and less flexibility to recover from wasted time, money, or energy. As a result, short-trip planning is not only a logistical matter, but also a meaningful interaction design problem: users must make quick decisions under uncertainty, with incomplete knowledge of an unfamiliar city and limited resources available.⁶

2.2 Target Segment and Observable Behaviors

Our target users are young independent, budget-conscious travelers taking short leisure trips to unfamiliar cities, either alone or with a small group of friends. This segment is relevant because Gen Z and similar young traveler groups are highly connected, use smartphones extensively during travel, are strongly influenced by social media, and are highly attentive to value for money when planning trips.^{3,4,11}

Budget is not simply one preference among others for these users; it is often a central constraint that shapes the entire planning process. Reports on young travelers suggest that value for money is a major determinant of destination and activity choice, especially when trips are self-funded or organized with friends. In addition, short city breaks are particularly attractive for this group, which makes this context both relevant and realistic for the project.^{3,4}

Their planning behavior is also strongly influenced by social media and peer-generated content. Platforms such as TikTok, Instagram, and similar channels often act as sources of inspiration because

they quickly communicate visual atmosphere, trendiness, and social proof. At the same time, these travelers still rely on more traditional tools such as Google Maps, review platforms, and blogs when they need practical details such as opening hours, location, prices, or reviews.^{5,9}

When traveling with friends, planning is not only individual but collaborative. Travelers exchange links, screenshots, short videos, and reviews, negotiate preferences, and adjust plans while already on site. This means the planning process is not static. It evolves as new information appears, as energy levels change, and as group preferences need to be reconciled.⁷

Overall, this target segment is characterized by five recurring features: high digital dependence, strong budget sensitivity, attraction to short city trips, reliance on multiple online sources, and the need to coordinate decisions around changing contextual factors such as mood, energy, time, and social atmosphere.

2.3 Pain Point 1: Popular Options Are Not Necessarily Personally Suitable

A first structural problem is that mainstream travel tools tend to surface what is popular, prominent, or highly rated, rather than what is personally suitable in a specific moment. Google explains that local search results are mainly influenced by relevance, distance, and prominence, while Tripadvisor states that its Popularity Ranking is based on the quality, recency, and quantity of reviews.^{1,2}

These signals are useful because they reduce uncertainty at a general level: they help users identify places that are well known, frequently visited, or broadly appreciated. However, they do not directly answer the more personal question that matters in short-trip contexts: *is this right for me right now?* A highly rated restaurant, viewpoint, or nightlife venue may still be too expensive, too crowded, too tiring to reach, too touristy, or not aligned with the traveler’s current mood and desired vibe.

This mismatch matters because the value of a recommendation is not only determined by its average quality, but also by its contextual fit. A traveler who is tired after a long day may prefer a calm, inexpensive, and nearby place, whereas the same traveler in a different moment may actively seek a lively and social experience. In that sense, personal suitability depends on situational variables such as energy, budget, weather, group composition, and social intention, not just on collective approval or reputation.⁶

Research on recommendation bias also strengthens this point. Popularity bias suggests that already popular options tend to gain even more visibility over time, which can reinforce the same places repeatedly and reduce the diversity of what users see.¹⁰ As a result, users may be guided toward what is broadly validated by others, but not necessarily toward what best matches their own present state.

For short urban trips, this gap between popularity and personal fit becomes more costly. Because time and energy are limited, travelers have less room to experiment. A choice that is only “generally good” but poorly matched to the current situation may still feel like a bad decision in practice. Therefore, the problem is not simply that platforms show popular places; it is that they offer limited support for judging whether a popular place is personally appropriate in context.

2.4 Pain Point 2: Travel Planning Information Is Fragmented

A second major pain point is that travel planning information is fragmented across too many tools and channels. Research describes online travel planning as increasingly complex, non-linear, and distributed across different stages of the trip.⁶ Rather than following one simple decision flow, users often jump repeatedly between platforms depending on the kind of information they need at a given moment.

This fragmentation is easy to observe in practice. One tool may help identify nearby places, another may communicate atmosphere through visuals, another may provide price comparisons, and another may offer longer user comments or local advice. Social media may inspire the idea of going somewhere, Google Maps may confirm whether it is close and open, and a review site may help

evaluate whether it is worth the cost. Each platform contributes a partial signal, but no platform necessarily helps users integrate these signals into one coherent recommendation.⁹

This creates a hidden workload for the user. The difficulty is not only finding information, but also comparing heterogeneous types of information: ratings, videos, photos, crowd impressions, price ranges, travel times, and opinions from different people. These signals are often incomplete and sometimes contradictory. A place may look exciting on social media but receive mixed reviews. A restaurant may be highly rated but too expensive for the group. A location may seem attractive online but be impractical once distance, waiting time, or weather are considered.

In group travel, fragmentation becomes even more demanding because information gathering also becomes a coordination task. Travelers need to share options, negotiate priorities, and resolve disagreements, often through messaging apps and informal exchanges.⁷ The planning burden therefore includes both cognitive work and social work: integrating information and aligning people.

For short trips, this fragmentation is especially problematic because it consumes time that could otherwise be spent enjoying the trip itself. Every switch between platforms adds friction. Every extra search adds uncertainty. Every unresolved contradiction delays the decision. In that sense, current tools do not necessarily fail because they lack information; they fail because they place the burden of integration almost entirely on the traveler.

2.5 Pain Point 3: Turning Suggestions Into Concrete Decisions

A third pain point is the difficulty of turning broad suggestions into concrete, timely decisions. Even when users find many possible places or activities, they still face the more difficult question of what to do *now*. This transition from having options to making a confident choice is where much of the friction of short-trip planning actually happens.

One reason is choice overload. In tourism research, a very large number of options has been shown to increase the likelihood that users make no choice at all.⁸ In other words, more options do not always improve decision quality. Instead, they can create hesitation, postponement, and regret. This is particularly relevant in digital travel contexts, where users can quickly accumulate long lists of saved restaurants, viewpoints, attractions, bars, and activities without becoming any more certain about what best fits the moment.

A second reason is informational overwhelm. When users must process large volumes of scattered information, decision-making becomes slower and more demanding.⁶ This can lead travelers to postpone decisions until they are already on site, at which point the context may be even less favorable: they may be tired, hungry, under time pressure, or trying to coordinate with others. Under these conditions, the cost of a poor decision increases further.

What makes this pain point particularly important is that short-trip decisions are highly contextual. A recommendation that sounds good in the abstract may be a poor choice in the actual situation. For example, a top-rated cocktail bar may be appealing in theory, but not if the group only has one hour left, one member is exhausted, and another is trying to stay within a strict budget. Current platforms often provide lists of options, but lists do not translate contextual constraints into concrete guidance.⁹

As a result, travelers are often left with many recommendations but little confidence. The unresolved question is not “what exists?” but rather “what best fits our present conditions and is worth spending our limited time and money on?” In short urban trips, where each decision has a relatively high opportunity cost, this unresolved step becomes a central source of stress and inefficiency.

2.6 Design Insight

Taken together, these three pain points reveal a deeper issue. The core problem in short urban trip planning is not a lack of information or even a lack of recommendations. Instead, it is a lack of confidence in contextual fit. Young travelers need support in assessing whether an option is suitable for their current state, not just whether it is popular, available, or visually appealing.

This insight links the three pain points directly. Popularity-based platforms do not sufficiently account for personal and situational differences. Fragmented tools force users to integrate many incomplete signals by themselves. Large numbers of suggestions then make it difficult to convert information into a confident, timely decision. When contextual fit remains unclear, travelers often fall back on what seems safest or most obvious, such as highly rated places, trending spots, or the opinion of the most dominant member of the group.

Therefore, the unmet need is not for more travel content, but for better support in judging whether a recommendation matches the traveler’s current mood, energy, budget, social intention, and time constraints. This is the key domain-level insight that should guide the design direction of the project.

3 Competitive Analysis

3.1 Why These Competitors Were Chosen

The selected competitors represent three common ways in which young travelers currently plan activities in unfamiliar cities: practical navigation and nearby discovery, popularity-based comparison, and social-media-driven inspiration. Together, they cover the main tools that users already rely on, but they also reflect the main limitations identified in the domain research: popularity does not necessarily mean personal fit, planning is fragmented across multiple platforms, and existing tools do not adequately support confident decisions based on mood, energy, budget, and desired vibe.

3.2 Competitor 1: Google Maps

Google Maps is one of the most widely used tools for in-the-moment travel decisions. It helps users find nearby places, check opening hours, estimate travel time, view ratings, and access practical information such as directions and price level. Its main strength is efficiency: it supports quick logistical decisions and reduces uncertainty about where a place is and how to reach it.

However, Google Maps mainly supports practical suitability rather than experiential suitability. It helps users answer questions such as *Is it nearby? Is it open? Is it well rated?*, but it gives limited support for more subjective questions such as *Does this place match my mood? Will the atmosphere feel calm, lively, social, or relaxed?* As a result, it is strong for logistics, but weaker for contextual fit and anticipated experience.

3.3 Competitor 2: Tripadvisor

Tripadvisor supports travel planning through rankings, reviews, and lists of popular attractions, restaurants, and activities. Its main strength is that it helps users compare many options and quickly identify places that are broadly appreciated by other travelers. This is useful especially for first-time visitors who want a general overview of what is considered worth doing in a city.

However, Tripadvisor is largely oriented toward aggregate popularity and mainstream tourist appeal. It is less useful when users need recommendations that reflect short-term contextual factors such as low energy, limited time, strict budget, or a specific social intention. In other words, it helps users identify what is generally well regarded, but not necessarily what is best for them in a particular moment.

3.4 Competitor 3: TikTok

TikTok is increasingly used by young travelers as a source of travel inspiration and place discovery. Its main strength is that it communicates atmosphere quickly and vividly through short videos. Compared with classic review platforms, it is much better at helping users imagine the vibe of a place, such as whether it feels trendy, crowded, aesthetic, casual, or fun.

However, TikTok is weak as a structured decision-making tool. Content is trend-driven, hard to filter systematically, and often lacks practical details such as opening hours, price, distance, wait

time, or suitability for a specific budget and energy level. It is therefore useful for discovering ideas and sensing atmosphere, but much less effective for making a confident and efficient final decision.

3.5 Competitive Analysis Map

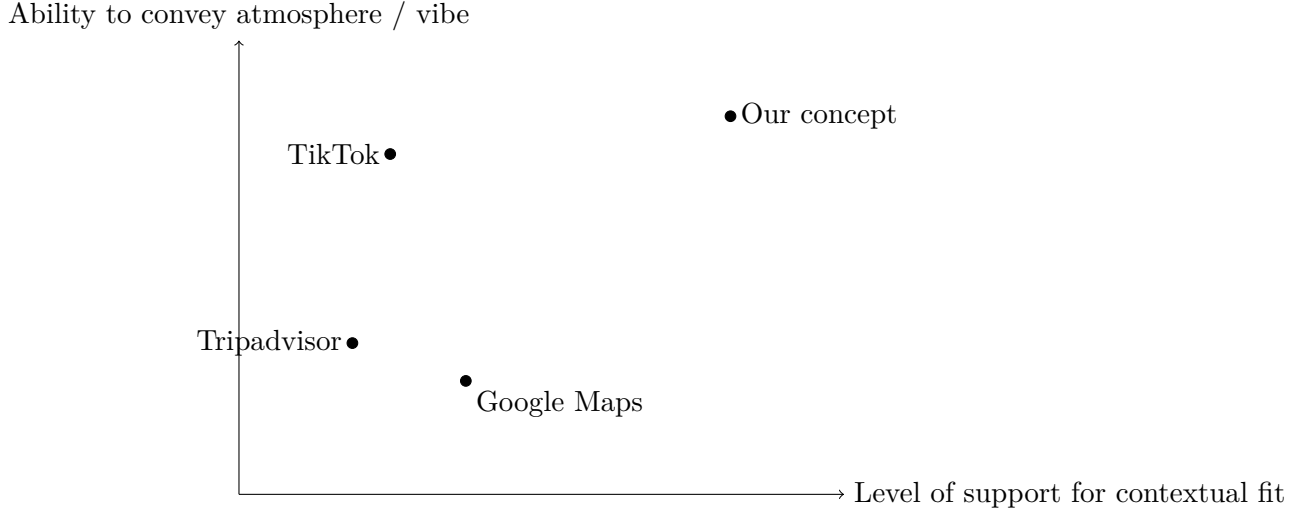


Figure 1: Competitive Analysis Map

Google Maps is positioned low-to-medium on contextual fit and low on atmosphere / vibe. Tripadvisor is low on contextual fit and low-to-medium on atmosphere / vibe. TikTok is high on atmosphere / vibe but low on contextual fit. Our concept is positioned in the top-right area because it aims to support both contextual fit and anticipation of atmosphere.

3.6 Our Position in the Competitive Landscape

Our concept is positioned in the top-right area of the CA map because it aims to combine two qualities that current tools rarely combine well. First, it would help users evaluate whether an option fits their current context, including mood, energy, budget, time constraints, and social intention. Second, it would help users anticipate the kind of atmosphere and experience a place is likely to offer. In this way, the concept does not simply provide more recommendations, but supports more confident decisions for short urban trips.

4 Design Problem Statement (DPS)

Young independent, budget-conscious travelers on short leisure trips in unfamiliar cities need a way to make spontaneous activity decisions without feeling that they are wasting limited time, money, or energy, because the real difficulty is not finding more recommendations, but judging whether an option fits their present context. When that contextual fit is unclear, they fall back on popularity cues or group opinion, which increases overload, hesitation, and regret.^{6,8,9}

5 SUNA Analysis

5.1 Suitability

This concept is suitable for an interaction design project because the challenge is not simply to collect travel information, but to design how users express changing needs and how the system turns them into clear, low-effort decisions. The main difficulty lies in supporting dynamic factors

such as mood, energy, budget, time, and desired social atmosphere without increasing cognitive load.

5.2 Usefulness

The concept is useful because it addresses a clear unmet need identified in the domain research. Existing tools help users find popular places, practical information, or inspiring content, but they do not adequately help them judge whether an option is personally suitable in the moment. This is especially valuable for short city trips, where limited time, money, and energy make poor decisions more costly.^{6,8,9}

5.3 Novelty

The novelty of the concept does not lie simply in using AI, but in how the system would translate changing situational factors into concrete travel decisions. Existing tools are usually strong in only one dimension: map and review platforms support logistics and comparison, while social platforms help users sense atmosphere and trendiness. Our concept is novel because it aims to combine these dimensions in one interaction flow by taking inputs such as mood, energy, budget, time constraints, and social intention, and turning them into recommendations that are both practically feasible and experientially appropriate. In this sense, the novelty lies in supporting contextual fit as an active decision variable rather than leaving users to infer it from fragmented signals across multiple platforms.

5.4 Adequacy

The assumed technology is appropriate for the target population because young travelers are highly digital, strongly influenced by social media, and increasingly comfortable using smartphones and AI tools for travel planning. A lightweight, mobile, context-aware tool therefore fits both their habits and the practical constraints of short-trip decision-making.^{3,4,11}

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